Landmark XVI-N — Scientific manuscript

Classical completion — PreparedClassicalBVInterfaceThroughOrderKappa
Science volume only (r12)

XVI-N r12 science volume

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Science volume. This PDF is the *scientific manuscript only* for Landmark XVI-N (revision r12). It contains Part I of the combined package: the body and the scientific notes it inputs (support lemma, massless-obstruction hypotheses, on-shell admission, field-only projection, mass-dependence / small-mass estimates, scaling audit, etc.). It does *not* include computational listings, note-source archives, or provenance manuscript dumps—those live in the companion appendix volume `XVI-N-appendix.pdf` and in the combined `XVI-N.pdf`.

Status (unchanged). Massless obstruction is conditional (Route A); Route B and arbitrary prescribed $m > 0$ remain unresolved; certified small-mass interval $0 \le m < 1.3715 \times 10^{-4}$.

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Abstract

Phase I constructs the unprepared classical BV complex and its prepared/switch deformations through $O(\kappa)$, with reconciled odd-field antibracket couplings, the antifield rule $s\Phi^*_A = (S, \Phi^*_A)$, and the invariant baseline $S_{\rm inv} = S_D - V_\sigma - V_g$. Gate N-Ins (Option B) uses the actual predecessor kernel $t_{\rm D}$ of XVI-I/XVI-H/XII-D as inherited by XVI-K. This revision separates the plain divergence $D_{\rm div}$ from the gauge-fixing divergence $D_{\rm gf} = D_{\rm div}\mathcal{P}$; the corrected root current is $H = \alpha + 2\mathcal{P}D^*_{\rm div}E^{\rm A}(D_{\rm div}\alpha)$, with vector $w = E^{\rm A}(D_{\rm div}\alpha)$ (not $E^{\rm A}(D_{\rm div}\mathcal{P}\alpha)$), certified exactly. All values computed for the superseded vector are withdrawn or recomputed. For the special family in the $m = 0$ instance of the frozen model we give admissible C^∞ data with $H|_U = 2m_0(e_0e_1 + e_1e_0)$, $\mathcal{C}|_U = 4m_0p'(t-z)\,dx^2$, $m_0 = \int S > 0$, and hence an analytic proof that $s_0F_t \neq 0$, plus a validated enclosure $s_0F_t \in [3.3566, 3.3637] \times 10^{-4}$ for an explicit compact ghost. Status: analytic 4D nonvanishing for the massless special family, conditional on the inherited reduction identities and configuration-space admissibility (support lemma discharged; Route A sufficient; Route B not claimed discharged). Conditional massless obstruction: Theorem 11. Fixed $m > 0$: certified small-mass interval $0 \le m < 1.3715 \times 10^{-4}$ (with $C_{\rm hi} \le 17843$; see Sec. 9.7). The antifield-linear completion (N-Ins.4) remains Unresolved as a universal statement.

Status.
N-A: **Constructed** (antifield prose, invariant/switched baseline and $D_{\rm div}/D_{\rm gf}$ conventions corrected and regression-verified).
N-Ins.1,2,5: Constructed. N-Ins.3: analytic 4D nonvanishing for the massless special family, conditional on the inherited reduction identities and configuration-space admissibility (Lem. 8; Route A sufficient; Route B not claimed discharged; Prop. 9; Thm. 11). Fixed $m > 0$: Sec. 9.7. N-Ins.4: Unresolved (universal).
1+1: exact nonzero variation in a 1+1 planar auxiliary model; no frozen-4D completion obstruction established.
Toy witness, inadmissible $(g-1)$ value, and all r7 values computed for $E^{\rm A}(D_{\rm div}\mathcal{P}\alpha)$: withdrawn.
N-B–N-E: not claimed.

Conventions: $\eta = \mathrm{diag}(1,-1,-1,-1)$, $q = \eta + \kappa h$, $a = \frac{1}{2}$, $\mathcal{P}$ = trace reversal. Frozen sheet: `conventions/convention_sheet.md`. External hashes: `provenance/RELEASE_MANIFEST.txt`.

1 Three classical objects

A broken master identity is not a BV complex: $Q_D^2F = (B_{\rm tot}, F)$ with $B_{\rm tot} = \frac{1}{2}(S_D, S_D)$.

(i) Unprepared / invariant $S_{\rm inv}$ with nilpotent $Q_{\rm inv}$;

(ii) nilpotent diffeomorphism BRST $s = s_0 + \kappa s_1$;

(iii) prepared/switched $S_D = S_{\rm inv} + V_\sigma + V_g$ with broken square.

(Equivalently one may report (S,S) without the $\frac{1}{2}$ and absorb conventions into B; we follow XIV-A in writing $(S,S) = \kappa\int c^\nu B_\nu + \cdots$ and stating $Q_D^2F = (B_{\rm tot}, F)$ with $B_{\rm tot} = \frac{1}{2}(S,S)$.)

2 Frozen convention block

Definition 1 (Antibracket)**.**

$$(F,G) = \sum_A \Big(\tfrac{\delta^R F}{\delta\Phi^A}\tfrac{\delta^L G}{\delta\Phi^*_A} - \tfrac{\delta^R F}{\delta\Phi^*_A}\tfrac{\delta^L G}{\delta\Phi^A} \Big), \quad QF := (S,F).$$

Then $(S,\Phi^A) = -\delta^R S/\delta\Phi^*_A$. *The blanket* $(S,\Phi) = \delta^L S/\delta\Phi^*$ *is false for odd* Φ.

Ordered couplings with $(S,\Phi^A) = s\Phi^A$ on every generator (source of truth: `conventions/source_of_truth.py`):

$$\boxed{S_{\rm af} = \int \big(h^{*\mu\nu}(sh)_{\mu\nu} + \phi^*(s\phi) - c^*_\mu(sc^\mu) - \bar c^{*\mu}(s\bar c_\mu) \big).} \tag{1}$$

Expanded (same source; signs locked to the box):

$$S_{\rm af} = \int h^{*\mu\nu}\big(R_0(c)_{\mu\nu} + \kappa a(\mathcal{L}_c h)_{\mu\nu}\big) + \kappa a \int \phi^*(c^\rho\partial_\rho\phi) \\ - \kappa a \int c^*_\mu\,(c^\rho\partial_\rho c^\mu) - i\int \bar{c}^{*\mu} b_\mu. \tag{2}$$

Antifield rule (not the reverse bracket):

$$\boxed{s\Phi^*_A = (S, \Phi^*_A).} \tag{3}$$

Generator+antifield tests and Q^2 on $S_{\rm inv}$ at retained order: `generator_antibracket_certificates.json`.

Common normalization. Set $a = \frac{1}{2}$ and $\xi^\mu = a\,c^\mu$. Parentheses symmetrization includes $\frac{1}{2}$: $\partial_{(\mu}c_{\nu)} := \frac{1}{2}(\partial_\mu c_\nu + \partial_\nu c_\mu)$. Define $R_0(c)_{\mu\nu} := \partial_{(\mu}c_{\nu)}$. The predecessor gauge generator of XI-A/XII-C, $(R_0^{\rm pred}\xi)_{\mu\nu} := \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, is renamed $R_0^{\rm pred}$; the letter K is reserved for the kinetic operator $-\Box\mathcal{P}$. Then $R_0(c) = R_0^{\rm pred}(ac)$. The letter K denotes only the kinetic operator $-\Box\mathcal{P}$ on tensors; never write $K\xi$ for the gauge map — use $R_0^{\rm pred}(\xi)$.

Displayed transformations.

$$s_0 h_{\mu\nu} = R_0(c)_{\mu\nu}, \qquad s_1 h_{\mu\nu} = a(\mathcal{L}_c h)_{\mu\nu}, \tag{4}$$
$$s_0\phi = 0, \qquad s_1\phi = a\,c^\rho\partial_\rho\phi, \tag{5}$$
$$s_0 c^\mu = 0, \qquad s_1 c^\mu = a\,c^\rho\partial_\rho c^\mu, \tag{6}$$
$$s_0\bar{c}_\mu = i b_\mu, \qquad s_1\bar{c}_\mu = 0, \tag{7}$$
$$s_0 b_\mu = 0, \qquad s_1 b_\mu = 0. \tag{8}$$

Equivalently $sh = a\mathcal{L}_c q$, $sq = \kappa a\mathcal{L}_c q$. The vector generating q is κac. Antifield transformations are induced by the extended action via $s\Phi^*_A = (S, \Phi^*_A)$, consistent with $Q = (S, \cdot)$ and the boxed rule above (left derivatives on odd variables). External $\sigma_\varepsilon, \chi_{\rm I}, g, t_{\rm D}$ are s-inert (fixed-external-data).

Gate 2 (BRST nilpotence)**.** $s_0^2 = 0$ *on generators (XI-B lem:nilpotence; written derivation). Interacting* $s^2 = 0$ *on unrestricted rational 2-jets: reported computation, with negative controls (wrong a, wrong ghost sign) failing as required (*`classical_bv/brst_normalization_certificates.json`*).*

Symmetric pairing: projector $(\Pi T)_{\mu\nu} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$, or $\sum_{\mu\le\nu}(2-\delta_{\mu\nu})h^{*\mu\nu}(sh)_{\mu\nu}$.

3 Extended actions

Definition 3 ($S^{[1]}_{\rm inv}$ and $S^{[1]}_D$)**.**

$$S^{[1]}_{\rm inv} = S^{[2]}_{\rm EH} + S^{[3]}_{\rm EH} + S^{\rm sc}_{\rm inv} + S_{\rm af}, \tag{9}$$
$$V_\sigma = \kappa\int(\sigma_\varepsilon - 1)\ell_3[h], \qquad V_g = \textit{switched-scalar deformation}, \tag{10}$$
$$S^{[1]}_D = S^{[1]}_{\rm inv} + V_\sigma + V_g. \tag{11}$$

Displayed antifield coupling: identical to (2) *(emitted from* `conventions/source_of_truth.py`*; regression asserts match). Conventions:*

- *left derivatives on odd variables;*
- *antifield pairing on all sixteen* $h_{\mu\nu}$ *components (not a symmetry-reduced 10);*
- $S'_{\ge 2} = 0$ *at this truncation (linear in antifields);*

- *unprepared / invariant theory:* $\boxed{S_{\rm inv} = S_D - V_\sigma - V_g.}$ *Setting* $\sigma_\varepsilon \equiv 1$ *removes only* V_σ *and gives the* switched *theory* $S_D - V_\sigma = S_{\rm inv} + V_g$ *(whose square still carries* $\mathcal{B}_g$*), not* $S_{\rm inv}$.

$S^{[2]}_{\rm EH}$ *is the Fierz–Pauli / quadratic EH density (XI-A* $S_{\rm FP}$*).* $S^{[3]}_{\rm EH} = \kappa \int \ell_3$ *is the coefficient of* κ *in* $-\frac{2}{\kappa^2}\int \sqrt{-q}\, R$*. Truncation:* $(S_{\rm inv}, S_{\rm inv}) = O(\kappa^2)$.

4 Cubic density ℓ_3 and Euler source J_3

Quote XII-C eq:ell3 / eq:J3:

$$\boxed{\ell_3 = -2(r_3 + b_1 r_2 + b_2 r_1).} \tag{12}$$

Here $r_n = \sum_{i=0}^{n-1} u_i^{\mu\nu} R_{n-i,\mu\nu}$ from the finite curvature expansion of $\sqrt{-q}R$ in powers of h, with $u_0 = \eta$, $u_1 = -h$ (raised), $u_2 = h \cdot h$, and $b_1 = \frac{1}{2}\tau$, $b_2 = \frac{1}{8}\tau^2 - \frac{1}{4}h_{\mu\nu}h^{\mu\nu}$, $\tau = \mathrm{tr}_\eta h$.

With

$$V_{1,\mu\nu} = R_{1,\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu} r_1, \qquad V_{2,\mu\nu} = R_{2,\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu} r_2 - \tfrac{1}{2} h_{\mu\nu} r_1,$$
$$J_3^{\mu\nu} = 2\Big(V_2^{\mu\nu} + b_1 V_1^{\mu\nu} - h^{\mu\alpha} V_{1\alpha}{}^{\nu} - h^{\nu\alpha} V_{1\alpha}{}^{\mu}\Big). \tag{13}$$

Compact variation: $\mathrm{Euler}_h(\ell_3) = J_3$ (XI-A sign). Implementation: `prep/grav_density.py`.

Weighted Euler of the prepared cubic:

$$\mathrm{Euler}_h(\sigma_\varepsilon \ell_3) = \sigma_\varepsilon J_3 + \mathcal{D}_{\sigma_\varepsilon}, \tag{14}$$

with time-only ramp (XII-C) $\mathcal{D}^{\mu\nu} = 2C_2^{\mu\nu 00}\sigma_\varepsilon'' - 2M_2^{\mu\nu 0}\sigma_\varepsilon'$. Define $Z := wJ_3 + \mathcal{D}$ with $w = \sigma_\varepsilon - 1$, so $\mathrm{Euler}_h\big(\ell_3 + w\ell_3\big) = J_3 + Z$ on the prepared deformation. $B_\nu := N_{0,\nu}(Z)$.

5 XII-C operators A, Q_ν, $N_{0,\nu}$, $\mathfrak{R}$

From XII-C eq:Noperators:

$$N_{0,\nu}(v) = 2\eta_{\rho\nu}\partial_\mu v^{\mu\rho}, \tag{15}$$
$$Q_\nu(v) = 2\partial_\mu(v^{\mu\rho}h_{\rho\nu}) - v^{\mu\rho}\partial_\nu h_{\mu\rho}, \tag{16}$$
$$\mathcal{N}_\nu(v) = N_{0,\nu}(v) + \kappa Q_\nu(v). \tag{17}$$

$A = 2G_{\rm lin}$ is twice the linearized Einstein operator in trace-reversed form (XI-A eq:Glin):

$$G^{\mu\nu}_{\rm lin} = -\tfrac{1}{2}\Box\bar{h}^{\mu\nu} + \partial^{(\mu}D^{\nu)} - \tfrac{1}{2}\eta^{\mu\nu}\partial_\rho D^\rho, \quad \bar{h} = \mathcal{P}h,\ D^\nu = \partial_\mu \bar{h}^{\mu\nu}. \tag{18}$$

Lemma (XII-C lem:divergence), verified here on unrestricted jets:

$$N_{0,\nu}(J_3) + Q_\nu(Ah) = 0, \qquad N_{0,\nu}(A) = 0. \tag{19}$$

Ramp defect (XII-C eq:Rampdef / eq:Zdivergence):

$$N_{0,\nu}(Z) = -wQ_\nu(Ah) + \mathfrak{R}_{\sigma_\varepsilon,\nu}. \tag{20}$$

Thus the metric-ramp breaking density $B_\nu := N_{0,\nu}(Z)$ matches XII-C's $-wQ + \mathfrak{R}$ exactly.

6 Universal prepared Noether (metric ramp)

Gate 4 ($O(\kappa)$ prepared Noether on unrestricted retained jets)**.** *Imported XII-C thm:noether + convention conversion (`prep/general_derivation_af0.md`). Sampling: "exact-arithmetic sampling of unrestricted retained jets passed" (20/20 Dual+Rational packs + 6/6 polynomial seeds; `prep/af0_noether_certificates.json`) — regression corroboration only, not a universal-identity certificate. Method 1 IBP: machinery check only.*

Target identity (af0 piece):

$$B^{[1]}_\nu = -wQ_\nu(Ah) + \mathfrak{R}_{\sigma_\varepsilon,\nu} = N_{0,\nu}(Z).$$

7 Master square by polynomial degree in antifields

Proposition 5 (Master square through $O(\kappa)$)**.** *In the linear-antifield truncation,*

$$\boxed{(S_D, S_D) = \kappa \int c^\nu \big(\mathcal{B}_{g,\nu} - B_\nu\big) + O(\kappa^2).} \tag{21}$$

Here $B_\nu = N_{0,\nu}(Z) = -wQ_\nu(Ah) + \mathfrak{R}_{\sigma_\varepsilon,\nu}$ is the metric-ramp piece, and $\mathcal{B}_{g,\nu}$ is the switched-scalar / V_g breaking retained in the prepared/switched square (certificate: `insertion/invariant_baseline_certificate.tex`). $E_\phi \partial_\nu \phi$ cancels between metric and scalar variations; $\sigma_\varepsilon = 1$ removes V_σ but not *$\mathcal{B}_g$.*

Decomposition by **polynomial degree in antifields** (not antifield number — note c^* has antifield number two):

- deg0: action variation / af0 breaking — Constructed (Gate 4 + $\mathcal{B}_g$);
- deg1: closure ($s^2 = 0$) — vanishes by Gate 2;
- deg$\geq$ 2: absent by truncation $S'_{\geq 2} = 0$.

Connecting derivation: in the antifield-linear truncation, $(S,S)|_{\Phi^*=0} = 2sS_{\text{field}}$. With $\delta_\xi S = -\kappa \int \xi\, B$ and $\xi = ac = c/2$,

$$\boxed{(S,S) = -\kappa \int c^\nu B_\nu}$$

on the metric-ramp piece. Combined with the switched-scalar contribution this yields (21), i.e. $(S_D, S_D) = \kappa \int c(\mathcal{B}_g - B) + O(\kappa^2)$. Here $\mathcal{B}_g$ is retained in the *prepared/switched breaking*, not in the invariant baseline S_{inv}.

7.1 Gauge-fixing sequence

Sequence:

$$S_D^{\text{BV}} \;\longrightarrow\; S_{D,\Psi}^{\text{BV}} = S_D^{\text{BV}} + s\Psi \;\longrightarrow\; S_{D,\Psi}\big|_{\Phi^*=0}. \tag{22}$$

With gauge fermion

$$\Psi = -i \int \bar{c}_\mu \big(2D^\mu[h] - b^\mu\big), \tag{23}$$

graded Leibniz $s(\bar{c}X) = (s\bar{c})X - \bar{c}\, sX$ gives

$$\begin{aligned} s\Psi &= \int b_\mu (2D^\mu - b^\mu) + 2i \int \bar{c}_\mu D^\mu[sh] \\ &= \int b_\mu (2D^\mu - b^\mu) + i \int \bar{c}\,\Box c + 2i\kappa a \int \bar{c}_\mu D^\mu(\mathcal{L}_c h), \end{aligned} \tag{24}$$

using $s\bar{c} = ib$, $sb = 0$, and $D[R_0 c] = \frac{1}{2}\Box c$ (`s_psi_certificates.json`).

Anticanonical T_Ψ: $(S_{D,\Psi}, S_{D,\Psi}) = (S_D, S_D) + O(\kappa^2)$ (breaking antifield-independent; Ψ field-only). Keep b off shell; elimination of b is separate. After eliminating b one recovers $S_{\text{gf}} = \int D_\nu D^\nu$ and ghost kinetic $i \int \bar{c}\,\Box c$.

Vertex table. (Emitted from `conventions/source_of_truth.py`.)

ID	Order	Fields	Factor / density
$V2_{\text{grav}}$	κ^0	hh	$K = -\Box\mathcal{P}$
$V2_{\text{ghost}}$	κ^0	$\bar{c}\,c$	$i\Box$
$V3_{\text{grav,prep}}$	κ^1	hhh	$\kappa\, \sigma_\varepsilon \ell_3$
$V3_{\bar{c}ch}$	κ^1	$\bar{c}\,c\,h$	$\kappa \cdot 2ia \int \bar{c}\, D(\mathcal{L}_c h)$
$V_{\text{af},h}$	$\kappa^{0,1}$	$h^* c (+h)$	$h^*(R_0(c) + \kappa a \mathcal{L}_c h)$
$V_{\text{af},c}$	κ^1	$c^* cc$	$-\kappa a\, c^*(c \cdot \partial c)$
$V_{\text{af},\bar{c}}$	κ^0	$\bar{c}^* b$	$-i\, \bar{c}^* b$

Graph↔action vertex relation: **open** / non-claim. Free action, gauge fixing, $K = -\Box\mathcal{P}$, and propagator interface are normalization inputs (N-Norm$_{\text{free}}$).

8 Q_D^2

Proposition 6. *$Q_D^2 F = (B_{\text{tot}}, F)$ with $B_{\text{tot}} = \frac{1}{2}(S_D, S_D)$. At $O(\kappa)$, B_{tot} is determined by (21). Direct smear on c^*: `qd2_antifield_certificates.json`. The prepared theory is not a BV complex.*

9 Gate N-Ins — actual t_{D}

Gate 7 (N-Ins)**.** *ClassicalInsertionInterfaceThroughOrderKappa (Option B).*

9.1 Sector decision

Preserve $F_t = \langle t_{\text{D}}, h \otimes h \otimes \phi\rangle$; reinstate scalar. ($F_t|_{\phi=0} = 0$ rejects Option A.) Decision: `insertion/N_insertion_sector_decision.md`.

9.2 N-Ins.1 — baseline and completion space

Sec. 3; `insertion/completion_space.json`. Minimal sector $(h, \phi, c; h^*, \phi^*, c^*)$; $\mathcal{I}_t = F_t + A_0 + \kappa A_1$; $\text{gh}(U, V, W) = (1, 1, 2)$; seed $\mathcal{I}_t|_{\Phi^*=0} = F_t$ (independent of b); no inverse operators.

9.3 N-Ins.2 — actual t_{D}, E_F, J_t

Definition (quoted from predecessors). Primary constructive definition — XVI-I Proposition `prop:test`, eq:push (sha256 `5c99e57543cbe65134e28810b7de605e8ded2b267e633f069aec834cc534bd80`):

$$\boxed{t_{\text{D}} = \sum_{|\alpha|+|\beta|\le 2} \partial_1^\alpha \partial_2^\beta\, \iota_*(c_{\alpha\beta} B_{f,g}) \in \mathcal{E}'(M^3),} \tag{25}$$

with $\iota(y, x) = (y, y, x)$, $B_{f,g}$ the compact-root kernel (XII-D lem:root-kernel / XVI-H), and $c_{\alpha\beta}$ the source coefficient tensors from j_D (XVI-I eq:j-delta), $|\alpha| + |\beta| \le 2$. The tensors $c_{\alpha\beta}^{J_3}$ are *implicitly* specified by ℓ_3/J_3 and the source-kernel identity; they are **not explicitly expanded** (XVI-I prop:test).

Equivalently XVI-H Theorem `thm:trep` / eq:O-t (enter-companion copy sha256 `1552ea91d0df41bf88b1b8e7c341bfefedfc27441233e2e7fd28bfc7112e8489`):

$$\mathcal{O}_{\text{pol}}(a, c; \phi) = \langle t_{\text{D}},\, a \otimes c \otimes \phi\rangle.$$

Root form (XII-D thm:root / XVI-H eq:O-root), writing $a_{\text{root}} := a_{f,g}[\phi]$:

$$F_t = \langle t_{\text{D}}, h \otimes h \otimes \phi\rangle = \int (a_{\text{root}})_{\mu\nu}(y)\, \mathcal{J}_{\text{D}}^{\mu\nu}[h](y)\, \mathrm{d}^4 y. \tag{26}$$

XVI-K inheritance. XVI-K (sha256 `6264bc8ec68534b8486c418596139fd20610d7f49c2f09b5113171e0be0789a8`) retains XVI-J construction data and does *not* redefine t_{D} as a new closed-form kernel. It inherits: ambient $M = \mathbb{R}^4$; preparation f, g, σ_ε; independent compact switch $\chi_{\text{I}} \in C_c^\infty(M)$; polarized source $Q^{\mu\nu} = 2C_2^{\mu\nu\alpha\beta}\partial_\alpha\partial_\beta\sigma_\varepsilon$; metric-dual basis $B^i = (G^{-1})^{ii} B_i$ with $G = \text{diag}(1, -2, -2, -2, 1, 2, 2, 1, 2, 1)$; imported compact root $B_{f,g}$ under Hypothesis H$_{\text{root}}$:

(R1) the composition exists;

(R2) compact support;

(R3) $\text{WF}(B_{f,g}) \subset Z_2 = \{(x_1, p_1; x_2, p_2) : p_1 + p_2 = 0\} \setminus 0$.

Switch χ_{I} multiplies densities undifferentiated in the main attachment ($a_{pq} \propto \chi_{\text{I}}(w)\, \mathcal{K}_{pq}$); χ_{I}-derivatives appear only in ambiguity / Leibniz terms of XVI-K, not as a redefinition of t_{D}.

Properties of t_{D}.

- Compactly supported: yes ($\operatorname{supp} t_{\mathrm{D}} \subset \{y_1 = y_2 \in C_K\} \times O_{f,g}$);
- Tensor structure: first two slots = symmetric contravariant 2-tensors; third slot = scalar ϕ;
- Slot-symmetric in the two metric slots;
- $\mathrm{WF}(t_{\mathrm{D}}) \subset Z_3$ (XII-D thm:admission / XVI-I eq:t-bounds);
- Fully specified *constructively* via $B_{f,g}$, j_D, ι_*, but component tensors $c^{J_3}_{\alpha\beta}$ only through hypotheses / implicit polynomial — not a single numerically expanded kernel.

Provenance digest: `insertion/tD_definition.md`.

Functional derivatives and obstruction current. $s\, t_{\mathrm{D}} = 0$ (fixed-external-data). Polarizing both metric slots:

$$E_F^{\mu\nu}(x) = 2 \int (a_{\mathrm{root}})_{\alpha\beta}(y)\, \mathcal{J}_{\mathrm{D}}^{\alpha\beta}\big(\delta_x^{\mu\nu}, h\big)(y)\, \mathrm{d}^4 y \tag{27}$$

(factor 2 from slot exchange; distributional derivatives of t_{D} / of j_D enter through $\mathcal{J}_{\mathrm{D}}$). Then

$$J_t^\nu = \partial_\mu E_F^{\mu\nu}, \qquad \boxed{s_0 F_t = 2 \int (a_{\mathrm{root}})_{\mu\nu}\, \mathcal{J}_{\mathrm{D}}^{\mu\nu}(R_0(c), h)\, \mathrm{d}^4 y = -\int c_\nu J_t^\nu.} \tag{28}$$

Kernel derivatives: yes (through $\partial_1^\alpha \partial_2^\beta$ in (25)). χ_{I}-derivatives: not in the main t_{D} seed of XVI-I/XVI-H; appear only in XVI-K ambiguity terms (scoped out of F_t). Artifacts: `insertion/seed_functional_derivatives.tex`, `insertion/free_obstruction_current.tex`.

9.4 N-Ins.3 — on-shell test

Invariant free shell: $G_{\mathrm{lin}}(h) = 0$, $(\Box + m^2)\phi = 0$. Compact ghost $c \in \mathfrak{g}_a$.

Frozen support (quoted). XII-D/XI-B:

$$g \in C_c^\infty(M_T; [0,1]), \qquad \overline{U} \subset \mathrm{Int}\{g = 1\} \cap \{\varepsilon < t < T\}, \qquad \mathfrak{g}_a = C_c^\infty(U; TM).$$

Ramp $\sigma_\varepsilon(t) = \chi(t/\varepsilon)$ with $\chi = 1$ for $s \geq 1$, so on U one has $\sigma_\varepsilon = 1$ and $\sigma'_\varepsilon = \sigma''_\varepsilon = 0$ (hence $\mathcal{D}_\sigma = 0$ for $\xi \in \mathfrak{g}_a$). XII-D prop:causal-plateau adds the conditional $g = 1$ on a neighbourhood of $J^-(K) \cap \operatorname{supp}(K\rho_\xi)$. XI-D allows switches/plateaus to be *chosen* to contain a given observation region; it does not force $g = 1$ on every $\operatorname{supp} f$.

Reduction chain (hypotheses at each step). On the invariant free shell $Ah = 0$, with $\xi \in \mathfrak{g}_a$ so $\operatorname{supp} \mathcal{L}_\xi h \subseteq \operatorname{supp} \xi \subset U$, and with plateau $\sigma_\varepsilon = 1$ on U (justifying the drop of $\mathcal{D}_\sigma$ *in this pairing only*, not globally):

$$\begin{aligned}
s_0 F_t &= 2 \int a_{\mathrm{root}} : \mathcal{J}_{\mathrm{D}}(R_0(c), h) \quad \text{(XVI-I eq:s0Ft; fixed preparation)} \\
&= 2 \int a_{\mathrm{root}} : J_3^{\mathrm{pol}}(R_0^{\mathrm{pred}}(\xi), h) \quad (\sigma_\varepsilon = 1 \text{ on } U;\ \mathcal{D}_\sigma \text{ omitted in the pairing}) \\
&= -\langle a_{\mathrm{root}},\, A(\mathcal{L}_\xi h)\rangle \quad \text{(XI-E thm:cubicvar + 1/2-polarization; jet-checked)} \\
&= -\langle H,\, \mathcal{L}_\xi h\rangle, \qquad H := A a_{\mathrm{root}}, \quad \text{(A formally self-adjoint; IBP OK by compact } \xi \subset U).
\end{aligned} \tag{29}$$

XII-D prop:causal-plateau ($g = 1$ on a neighbourhood of $J^-(K) \cap \operatorname{supp}(K\rho_\xi)$) is an *extra* assumption, kept separate from the basic $g = 1$ on U.

Two divergence operators (r8 correction). With the natural pairings $\langle T, S\rangle = \int T^{\mu\nu} S_{\mu\nu}$, $\langle v, u\rangle = \int v^\nu u_\nu$ (indices moved with η, parentheses with $\frac{1}{2}$), define

$$D_{\rm div} T := \partial_\mu T^{\mu\nu}, \qquad D_{\rm gf} r := \partial_\mu (\mathcal{P} r)^{\mu\nu} = D_{\rm div} \mathcal{P} r, \qquad D^*_{\rm div} v = -\partial^{(\mu} v^{\nu)}, \qquad D^*_{\rm gf} = \mathcal{P} D^*_{\rm div}.$$

From the displayed $G_{\rm lin}$ (XI-A eq:Glin), with no change of pairing or variables,

$$A = 2G_{\rm lin} = -\Box\mathcal{P} - 2D^*_{\rm gf} D_{\rm gf} = -\Box\mathcal{P} - 2\mathcal{P} D^*_{\rm div} D_{\rm div} \mathcal{P}. \tag{30}$$

For $a_{\rm root} = -E^{\rm A}\mathcal{P}\alpha$ ($\vartheta = 1$), using $\mathcal{P}^2 = 1$, $\Box E^{\rm A} = 1$ and commutation of constant-coefficient operators with $E^{\rm A}$ on compactly supported data,

$$\boxed{H = A a_{\rm root} = \alpha + 2D^*_{\rm gf} E^{\rm A} D_{\rm div}\alpha = \alpha + 2\mathcal{P} D^*_{\rm div} w, \qquad w = E^{\rm A}(D_{\rm div}\alpha).} \tag{31}$$

The r7 text used D for both operators and the vector $E^{\rm A}(D_{\rm div}\mathcal{P}\alpha)$; this is **wrong** for traceful α (the two differ by $\frac{1}{2}\partial^\nu$ of a trace) and is superseded. Independent exact certificate (`insertion/root_adjoint_Ddiv_Dgf_check.py`): adjointness of $D^*_{\rm div}$, $D^*_{\rm gf}$ up to explicit total divergences; (30) component by component on ten generic functions; $H_{\rm direct} = 2G_{\rm lin}(a_{\rm root})$ equals (31) for generic traceful tensors and for explicit pure-trace, mixed and trace-free polynomial tensors, while the superseded formula fails for every traceful case and agrees only for trace-free ones; and $E^{\rm A}\partial_\lambda = \partial_\lambda E^{\rm A}$ exactly on the domain used (advanced Kirchhoff cone integrals of sources $S^{(k)}(t)\,\mathrm{poly}(x)$, 24 cases). Since $D_{\rm div} A = 0$ (Bianchi), $D_{\rm div} H = 0$.

(a) Special family. XII-D eq:alpha (quoted): $\alpha_{\rm co} = -\sigma\, \mathfrak{a}^{\lambda\mu,ab}\partial_\lambda E^{\rm A}(f\partial_\mu\phi)$, $\alpha_{\rm sc} = g\, \mathfrak{a}^{\mu\nu,ab}\partial_\mu v_f \partial_\nu\phi - \frac{m^2}{2} g\eta^{ab} v_f \phi$ (raw factors σ, g). For $m = 0$, $\phi = x^1$, $E^{\rm A}(f\partial\phi) = (\partial\phi) v_f$:

$$\alpha^{ab} = (g-\sigma)\, \mathfrak{a}^{\mu 1,ab}\, \partial_\mu v_f, \qquad (D_{\rm div}\alpha)^b = \partial_a (g-\sigma)\, \mathfrak{a}^{\mu 1,ab}\partial_\mu v_f - \tfrac{1}{2}(g-\sigma)\eta^{b1} f$$

($\Box v_f = f$), both verified symbolically for generic g, σ, v_f (`insertion/nins3_4D_special_family_analytic.py`). Hence α vanishes on the plateau $\mathrm{Int}\{g = \sigma = 1\}$, and $\mathrm{supp}\, D_{\rm div}\alpha \subset (\mathrm{supp}\, d(g-\sigma) \cap \mathrm{supp}\, dv_f) \cup (\mathrm{supp}(g-\sigma) \cap \mathrm{supp}\, f)$: the ramp, plus the part of $\mathrm{supp}\, f$ lying off the plateau. It is ramp-only when $\mathrm{supp}\, f \subset \mathrm{Int}\{g = \sigma\}$. On U: $H|_U = 2\mathcal{P} D^*_{\rm div} w$.

(b) General family. For generic independent jets $\alpha|_U \not\equiv 0$ and $D_{\rm div}\alpha|_U \not\equiv 0$ (`insertion/root_adjoint_H.py`). No plateau-vanishing or ramp-only support statement is made for the general family.

Mass in the frozen contract. XI-A/XI-B/XII-C fix $P_{\rm free} = \Box + m^2$ with $m \geq 0$, the mass being fixed for each instance of the model. So $m = 0$ is a permitted instance, and every massless result below is a result for the $m = 0$ instance only. For a fixed $m > 0$ instance it is a model restriction, not a witness.

Root current $\mathcal{C}$. With $(\mathcal{L}_\xi h)_{\mu\nu} = \xi^\rho \partial_\rho h_{\mu\nu} + h_{\rho\nu}\partial_\mu \xi^\rho + h_{\mu\rho}\partial_\nu \xi^\rho$ (linear in h; the κ^0 term $R_0\xi$ is $s_0 h$ and is not part of $\mathcal{L}_\xi h$), the reduction chain gives $s_0 F_t = -\langle H, \mathcal{L}_\xi h\rangle$. For symmetric H,

$$-H^{\mu\nu}(\mathcal{L}_\xi h)_{\mu\nu} = \mathcal{C}_\rho \xi^\rho - \partial_\mu\big(2H^{\mu\nu} h_{\rho\nu}\xi^\rho\big), \qquad \boxed{\mathcal{C}_\rho = 2\partial_\mu(H^{\mu\nu} h_{\rho\nu}) - H^{\mu\nu}\partial_\rho h_{\mu\nu}} \tag{32}$$

(checked symbolically for generic H, h, ξ; $\mathcal{C} = Q(H)$ in XII-C notation), so $s_0 F_t = \langle \mathcal{C}_\rho, \xi^\rho\rangle$ for $\xi \in \mathfrak{g}_a$. This is a distributional identity of the pairing. Adding $R_0\xi$ to $\mathcal{L}_\xi h$ would not change it, since $AR_0 = 0$ and $D_{\rm div} H = 0$. The cubicvar step $2J_3^{\rm pol}(R_0^{\rm pred}\xi, h) = -A(\mathcal{L}_\xi h)$ on $Ah = 0$ was re-checked pointwise with the actual J_3 (XII-C eq:J3) for random cubic ξ. A control dropping the transport term fails. Equivalently $s_0 F_t = -2\langle w, D_{\rm gf}(\mathcal{L}_\xi h)\rangle$ on U.

9.5 Admissible 4D special-family computation ($m = 0$ instance)

Lemma 8 (Actual-source support)**.** *Let $T = 4$, $\varepsilon = \frac{1}{2}$, $U = (0.9, 1.1) \times (-0.1, 0.1)^3$, and let χ be the XII-D ramp profile. Define*

$$\begin{aligned} S(u) &= b(2u-5), \qquad b(v) = e^{1-1/(1-v^2)}\ (|v|<1),\ 0 \text{ else}, \\ \beta(t) &= \chi\big((t-1.8)/0.1\big)\chi\big((3.2-t)/0.1\big), \\ \lambda(r) &= \chi\big((r-0.6)/0.1\big)\chi\big((2.5-r)/0.2\big), \\ \kappa_0(t) &= \chi\big((t-0.55)/0.05\big)\chi\big((3.6-t)/0.1\big), \\ \mu(r) &= \chi\big((3.2-r)/0.2\big), \end{aligned}$$

so $S \in C_c^\infty(\mathbb{R}; [0,1])$ with $\operatorname{supp} S = [2,3]$ (not claimed inside $(2,3)$), $\operatorname{supp}\beta = [1.8, 3.2]$ with $\beta = 1$ on $[1.9, 3.1]$, $\operatorname{supp}\lambda = [0.6, 2.5]$ with $\lambda = 1$ on $[0.7, 2.3]$, $\kappa_0 = 1$ on $[0.6, 3.5] \supset \operatorname{supp}\beta$, and $\mu = 1$ on $\{r \le 3\} \supset \operatorname{supp}\lambda$. Set $\psi = \beta(t)\lambda(r)r^2$, $f = \Box\psi$, $v_f = E_0^{\mathrm{A}} f = \psi$, $g = \kappa_0(t)\mu(r)(1 - S(t))$, and $\sigma_\varepsilon(t) = \chi(t/\varepsilon)$. Then:

1. *$g \in C_c^\infty(M_T; [0,1])$, $g = \sigma = 1$ on a neighbourhood of $\overline{U}$, and $\overline{U} \subset \operatorname{Int}\{g = 1\} \cap \{\varepsilon < t < T\}$.*

2. ***(Global support.)*** *On a neighbourhood of $\operatorname{supp} d\psi$ one has $\kappa_0 = \mu = \sigma = 1$, hence $g - \sigma = -S$. Outside $\operatorname{supp} d\psi$, the massless special-family identity $\alpha = (g - \sigma)\,\mathfrak{a}^{\mu 1}\partial_\mu v_f$ (XII-D eq:alpha at $m = 0$, $\phi = x^1$, $v_f = \psi$) forces $\alpha = 0$. Therefore*

$$\alpha = -S(t)\,\mathfrak{a}^{\mu 1}\partial_\mu\psi \qquad \text{globally on } M_T,$$

   and $\operatorname{supp}_t(D_{\mathrm{div}}\alpha) \subset [2,3]$.

3. *On the open set $N = \{t \in (1.9, 3.1)\} \times \{0.7 < r < 2.3\}$ one has $\beta = \lambda = \kappa_0 = \mu = \sigma = 1$ with all derivatives vanishing, hence $\psi = r^2$ and*

$$D_{\mathrm{div}}\alpha = (S'x,\ 3S,\ 0,\ 0)$$

   on N. Combined with item 2, $D_{\mathrm{div}}\alpha$ vanishes off the time slab $[2,3]$: temporal cutoff edges of β and the κ_0/μ transitions contribute no source; radial cutoff contributions of λ during $[2,3]$ do not meet the relevant advanced cones from $\overline{U}$ (item 4). (No claim is made that radial edges contribute no source globally.)

4. *Every advanced cone from a point of $\overline{U}$ meets $\{t \in \operatorname{supp} S\}$ only at radii in $[2 - 1.1, 3 - 0.9] = [0.9, 2.1]$, hence at spatial radii in $[0.726, 2.274] \subset (0.7, 2.3)$. Those cone points therefore lie in N. By items 2–3 the future cones from U see exactly the model source $D_{\mathrm{div}}\alpha = (S'x, 3S, 0, 0)$ (no exterior-N contribution).*

5. *The collar ϑ of XII-D eq:compact-root equals 1 on a neighbourhood of $[0,T]$, hence on U and on every cone point used above; insertion of ϑ changes nothing (XII-D thm:root). The independent switch χ_{I} of XVI-J/K does not occur in t_{D} (XVI-I eq:push / XVI-H thm:trep / XII-D $B_{f,g}$) and does not affect F_t.*

6. *Consequently the local root formulas on U,*

$$w = (-m_0 x,\ 3(m_1 - m_0 t),\ 0,\ 0), \qquad H^{01} = 2m_0,$$

   follow from the actual compact construction by 4D Huygens (cone support of E_0^{A}), not from polynomial surrogate sources.

Proof. Item 1 is an exact containment of the support intervals. For item 2: $\operatorname{supp} d\psi \subset \operatorname{supp}\beta \times \operatorname{supp}\lambda$ up to the radial factor r^2, and on that set $\kappa_0 = \mu = 1$ by construction while $\sigma = 1$ because $t \ge 1.8 > \varepsilon$; thus $g - \sigma = -S$ on a neighbourhood of $\operatorname{supp} d\psi$. Off $\operatorname{supp} d\psi$ one has $\partial\psi = 0$, so the special-family factorisation $\alpha = (g - \sigma)\mathfrak{a}^{\mu 1}\partial_\mu\psi$ yields $\alpha = 0$. The global formula and the time support of $D_{\mathrm{div}}\alpha$ follow. Item 3 is the symbolic reduction of XII-D eq:alpha on the plateau set N (`nins3_support_lemma_certificates.json`, Parts P1–P2), together with item 2 excluding all source outside the time slab $[2,3]$. Item 4 is the cone-radius

computation $R \in [2-1.1,\, 3-0.9]$ and $r \in [R-\sqrt{3}\cdot 0.1,\, R+\sqrt{3}\cdot 0.1] \subset (0.7, 2.3)$, so every contributing cone point lies in N where the model source holds. Item 5 quotes XII-D thm:root and the XVI-K inheritance recorded in Section 9.3. Item 6 is then the Kirchhoff evaluation of Proposition 9. Numeric corroboration (Part P3 of the certificate) reproduces $J_{\text{actual}} = J_{\text{model}}$ at random cone points to 3×10^{-21} and the closed forms for w on U to relative 10^{-14}; it is corroboration only. □

Data (all C^∞; summary of Lemma 8). $T = 4$, $\varepsilon = \frac{1}{2}$, $U = (0.9, 1.1) \times (-0.1, 0.1)^3$. $S \in C_c^\infty(\mathbb{R}; [0,1])$ with $\operatorname{supp} S = [2,3]$ (not claimed inside $(2,3)$), $S \not\equiv 0$, $m_k := \int u^k S(u)\, du$. $\psi = \beta(t)\lambda(r) r^2$ with the explicit cutoffs of Lemma 8 ($\beta = 1$ on $[1.9, 3.1]$, $\lambda = 1$ on $[0.7, 2.3]$); $f := \Box\psi \in C_c^\infty(M_T)$, so $v_f = E_0^{\mathrm{A}} f = \psi$. $g = \kappa_0(t)\mu(r)(1 - S(t))$ with $\kappa_0 = 1$ on $[0.6, 3.5] \supset \operatorname{supp}\beta$, $\mu = 1$ on $\{r \le 3\} \supset \operatorname{supp}\lambda$; then $g \in C_c^\infty(M_T; [0,1])$, $g = \sigma = 1$ near $\overline{U}$, and $\overline{U} \subset \operatorname{Int}\{g = 1\} \cap \{\varepsilon < t < T\}$. $\phi = x^1$, $m = 0$. $h_{12} = h_{21} = p(t-z)$ (TT plane wave, $G_{\text{lin}} h = 0$, radiative if $p'' \neq 0$); admitted by Route A of Section 9.6 (Route B not claimed discharged). By Lemma 8, every advanced cone from U sees exactly $J = D_{\text{div}}\alpha = (S'x,\, 3S,\, 0,\, 0)$, so w and a_{root} on U are exact cone integrals of that model source (no surrogate).

Proposition 9 (Special-family 4D nonvanishing, $m = 0$ instance)**.** *On U: $\alpha = 0$, $w = (-m_0 x,\, 3(m_1 - m_0 t),\, 0,\, 0)$, $a_{\text{root}}^{ab} = (m_1 - m_0 t)(\mathcal{P}\mathfrak{a}^{\mu 1}\partial_\mu r^2)^{ab}$, $H = 2G_{\text{lin}}(a_{\text{root}}) = 2\mathcal{P}D_{\text{div}}^* w = 2m_0(e_0 \otimes e_1 + e_1 \otimes e_0)$, $\mathcal{C} = 4m_0\, p'(t-z)\, dx^2$. Hence for every $\xi \in C_c^\infty(U; TM)$*

$$s_0 F_t = 4m_0 \int p'(t-z)\, \xi^2\, d^4x,$$

which is > 0 whenever $p' > 0$ on $\operatorname{supp}\xi$ and $0 \le \xi^2 \not\equiv 0$. In particular $\mathcal{C}|_U \neq 0$, and $\xi = e_2\chi$ with $\chi \ge 0$ detects it.

Proof. Exact symbolic computation (`insertion/nins3_4D_special_family_analytic.py`). Write $E^{\mathrm{A}} j(t,x) = \int_0^\infty R\, \langle j(t+R, x+Rn)\rangle_{S^2} dR$. Sphere averages of polynomials are exact, and $\int (u-t)^p S^{(k)}(u) du$ is reduced by parts to the moments m_i. H_{direct} from the displayed G_{lin} equals (31); $\Box w = \Box a_{\text{root}} = 0$ and $D_{\text{div}} H = 0$ on U. With $D_{\text{div}} H = 0$ and only $h_{12} \neq 0$, (32) gives $\mathcal{C}_2 = 2H^{01}\partial_0 h_{21} = 4m_0 p'$ and $\mathcal{C}_{0,1,3} = 0$. $m_0 > 0$ because $S \ge 0$, $S \not\equiv 0$. □

Complete original pairing. For $\xi = e_2 \prod_i ((X_i - \ell_i)(u_i - X_i))^3$ on the box $[0.95, 1.05] \times [-0.05, 0.05]^3$, the following coincide exactly: $-\langle a_{\text{root}}, A\mathcal{L}_\xi h\rangle$, $-\langle H, \mathcal{L}_\xi h\rangle$, $\langle \mathcal{C}, \xi\rangle$ and the closed form, all equal to $m_0(p_1 + 2p_2)/(9.604 \times 10^{35})$ for $p = p_1 u + p_2 u^2$. The complete original pairing $2\int a_{\text{root}} : J_3^{\text{pol}}(R_0^{\text{pred}}\xi, h)$ (actual J_3, Gauss–Legendre exact for the degree, 50 digits) reproduces it; relative deviation 2.5×10^{-39} at $p_1 = 1, p_2 = \frac{1}{2}$ (polynomial C^2 test ghost on a box in U); numerical corroboration only. This checks, to 50 digits, the full pairing $s_0 F_t = 2\int a_{\text{root}} : \mathcal{J}_{\mathrm{D}}(R_0(c), h)$ ($\sigma = 1$ on $\operatorname{supp}\xi$), not only the reduced surrogate.

Validated enclosure. Take $b(v) = e^{1 - 1/(1-v^2)}$ ($|v| < 1$), $S(u) = b(2u - 5)$ so $\operatorname{supp} S = [2,3]$ and $S \in C_c^\infty(\mathbb{R}; [0,1])$ (not claimed inside $(2,3)$), $\xi = e_2\, b(\frac{t-1}{d}) b(\frac{x}{d}) b(\frac{y}{d}) b(\frac{z}{d})$ with $d = 0.09$, and $p(u) = u$ (so $p' = 1$). Then $s_0 F_t = 2d^4 I_b^5$, $I_b = \int_{-1}^1 b$. Monotone interval Riemann sums with outward rounding (`mpmath.iv`) give

$$s_0 F_t \in [3.3566 \times 10^{-4},\, 3.3637 \times 10^{-4}], \qquad 0 \notin [L, U].$$

The support and regularity are exact by construction, and the root quantities are closed forms with no root error. The quadrature error is enclosed (`insertion/nins3_4D_validated_enclosure.py`). *Conditional on:* the inherited identities XVI-I eq:s0Ft and XI-E thm:cubicvar, the latter re-checked pointwise. *Scope:* special family, $m = 0$ instance. Configuration-space admission: Route A of Section 9.6 is discharged and sufficient; Route B is not claimed discharged (support defect recorded there).

Structural $O = D_{\text{gf}}\mathcal{L}.h$. For constant TT $h_{12} = 1$, $\mathcal{C} = 2h\, D_{\text{div}} H = 0$ (vanishing subcase). For free $w = t - x$ the exact pairing $\langle w, O\xi\rangle_U = -1/1296$ involves a *free* w. It is unaffected by the correction and is structural only.

Inadmissible $(g-1)$ on ghost support. Withdrawn (`admissibility_verdict.md`).

Superseded r7 values and recomputation. 1+1 **planar auxiliary model** (`insertion/nins3_1p1_corrected.py`, two region-split evaluations each). The r7 configuration ($h_{01} = t-x$, $s = -\psi$) reproduces the old 307/4054050 for the superseded source. With the corrected source its value is exactly 0 ($H^{00} = H^{11} = E^{\mathrm{A}}(\frac{1}{2}\Box(s\psi_x))|_U = 0$), so 307/4054050 **is withdrawn**. With a time-only ramp $s = -(t-2)(3-t)$ the corrected values are 2/4725 ($h_{01} = t - x$, ξ^0) and $-1891/4989600$ ($h_{22} = t$, ξ^0). Verdict: **Exact nonzero variation in a** 1+1 **planar auxiliary model; no frozen-**4**D completion obstruction established.** No lower-dimensional BV theory, seed or completion class is specified. Planar g is not compact in 4D, and the 1+1 Green function is not the 4D advanced kernel.

4**D Kirchhoff numerics (corroboration only).** The r7 values -2.913×10^{-3} (16×10) and -2.669×10^{-3} (12×8) were computed for the superseded vector. Their generating script is not in the artifact tree, so they are **withdrawn**. A reconstruction of the stated configuration (`insertion/nins3_4D_kirchhoff_corrected.py`) gives the corrected values -1.312×10^{-4} (16×10) and -1.277×10^{-4} (12×8), against -2.498×10^{-4} and -2.709×10^{-4} for the superseded functional: same sign, about half the size, as in Proposition 9 ($H = 2m_0$ against $H_{\mathrm{old}} = 4m_0$). This is not a certificate.

On-shell verdict. Special family, $m = 0$ instance: **Analytic** 4**D nonvanishing for the massless special family, conditional on the inherited reduction identities and configuration-space admissibility** (Prop. 9 + Lemma 8 + validated enclosure; hypotheses 9.6–9.6). The conditional massless obstruction is Theorem 11. Fixed $m > 0$: small-mass interval of Section 9.7 if the bound of Track B2 succeeds; otherwise Unresolved. General family: Unresolved.

9.6 N-Ins.4 — completion class and factorization scaffold

Declared class: *regular*, *seed-preserving*, *antifield-linear* $\mathcal{I}_t = F_t + A_0 + \kappa A_1$ in the Gate N-Ins kernel class, with

$$A_0 = \int \big(h^{*\mu\nu} U_{\mu\nu} + \phi^* V + c^*_\mu W^\mu\big), \quad \mathrm{gh}(U, V, W) = (1, 1, 2),$$

and *no new inverse operators* beyond those already in the inherited root construction. Degree-zero equation: $P_0 Q_0 A_0 = -s_0 F_t$. The field-only part of $Q_0 A_0$ is the Koszul–Tate image $\int(\delta S_0/\delta h)U + \int(\delta S_0/\delta\phi)V$, which lies in the free EOM ideal. A regular completion of this class can therefore cancel $s_0 F_t$ only if $s_0 F_t$ vanishes on shell. A nonzero admissible on-shell result would obstruct the declared regular, seed-preserving completion class through this equation-of-motion (Koszul–Tate) cancellation requirement. It would not do so through an automatic appeal to local cohomology, which does not apply to the multilocal kernel t_{D}. Proposition 9 supplies such an input for the $m = 0$ special family. Lemma 10 writes the EOM-ideal identity for the declared class; Theorem 11 records the resulting conditional obstruction for the constructed massless instance. N-Ins.4 remains **Unresolved** as a *universal* completion statement: the theorem covers one instance, the inherited reduction chain 9.6 is imported (re-checked, not re-proved), and fixed $m > 0$ is treated separately in Section 9.7. On U, $s_0 F_t = -2\langle w, D_{\mathrm{gf}}(\mathcal{L}_\xi h)\rangle$ with $w = E^{\mathrm{A}}(D_{\mathrm{div}}\alpha)$. Scaffold: `insertion/factorization_P0Q0A0.md`. No new inverse operators beyond those in the inherited root construction.

Hypotheses of the conditional massless obstruction.

(H1) Inherited seed. t_{D} is the distribution of XVI-I Proposition `prop:test`, eq:push (sha256 `5c99e57543cbe65134e28810b7de605e8ded2b267e633f069aec834cc534bd80`):

$$t_{\mathrm{D}} = \sum_{|\alpha|+|\beta|\le 2} \partial_1^\alpha \partial_2^\beta\, \iota_*(c_{\alpha\beta} B_{f,g}) \in \mathcal{E}'(M^3),$$

with $B_{f,g}$ the compact root of XII-D Theorem `thm:root` / Lemma `lem:root-kernel`, and $c_{\alpha\beta}$ the source coefficient tensors of XVI-I eq:j-delta. Equivalently XVI-H Theorem `thm:trep`. The seed functional is $F_t = \langle t_{\mathrm{D}}, h \otimes h \otimes \phi\rangle = \int a_{\mathrm{root}} : \mathcal{J}_{\mathrm{D}}[h]$ with $a_{\mathrm{root}} = a_{f,g}[\phi]$.

(H2) Reduction identities. On the invariant free shell $Ah = 0$, for $\xi \in \mathfrak{g}_a$ with $\operatorname{supp}\xi \subset U$ and with the plateau $\sigma_\varepsilon = 1$ on U (so that $\mathcal{D}_\sigma$ drops *in this pairing only*):

$$s_0 F_t = 2 \int a_{\mathrm{root}} : \mathcal{J}_{\mathrm{D}}(R_0(c), h) = -\langle a_{\mathrm{root}}, A\,\mathcal{L}_\xi h\rangle = \langle \mathcal{C}, \xi\rangle,$$

citing XVI-I eq:s0Ft and XI-E Theorem `thm:cubicvar` (the latter re-checked pointwise in this manuscript).

(H3) Admitted free configuration space. The free classical configurations are ambient smooth solutions of $Ah = 0$ and $(\Box + m^2)\phi = 0$ on a neighbourhood of the closed slab, with *no spatial-decay requirement* (XII-D D0: "Fields are ambient smooth configurations. No spatial decay is required."; XI-B: "No spatial decay was imposed on classical configurations."; XII-C: "Metric Cauchy data are unrestricted."). Compact ghosts $\xi \in \mathfrak{g}_a = C_c^\infty(U; TM)$ are required by the frozen support (XII-D/XI-B). The displayed free solutions $\phi = x^1$ and $h_{12} = p(t - z)$ are therefore admitted by Route A of Section 9.6. Route B (spatially compact Cauchy data) is **not** claimed discharged in this revision (support defect recorded there).

(H4) Completion class. Regular, seed-preserving, antifield-linear completions $\mathcal{I}_t = F_t + A_0 + \kappa A_1$ in the Gate N-Ins kernel class, with

$$A_0 = \int \big(h^{*\mu\nu} U_{\mu\nu} + \phi^* V + c_\mu^* W^\mu\big), \quad \mathrm{gh}(U, V, W) = (1, 1, 2),$$

and with no new inverse operators beyond those already in the inherited root construction (Gate N-Ins.1; `insertion/completion_space.json`).

(H5) Plateau and root-cutoff requirements. $g \in C_c^\infty(M_T; [0, 1])$, $\overline{U} \subset \mathrm{Int}\{g = 1\} \cap \{\varepsilon < t < T\}$, $\sigma_\varepsilon(t) = \chi(t/\varepsilon)$ with $\chi = 1$ for $s \geq 1$ (so $\sigma_\varepsilon = 1$ and $\sigma'_\varepsilon = \sigma''_\varepsilon = 0$ on U). The collar $\vartheta \in C_c^\infty((-\delta, T + \delta))$ of XII-D eq:compact-root equals 1 on a neighbourhood of $[0, T]$. The independent switch χ_{I} of XVI-J/K multiplies interaction densities outside F_t and does not enter t_{D}. Explicit support bounds for the special-family cutoffs $\beta, \lambda, \kappa_0, \mu$ are those of Lemma 8.

Scope. The theorem of Section 11 concerns **one specified kernel instance**: the massless special-family preparation of Proposition 9, with the explicit cutoffs of Lemma 8. An obstruction for that instance does not automatically establish obstruction for every inherited kernel, nor for a fixed $m > 0$ instance without further work (Track B).

Configuration-space admission.

Route A (quoted; discharged). XII-D D0: "Fields are ambient smooth configurations. No spatial decay is required." XI-B: "No spatial decay was imposed on classical configurations." XII-C: "Metric Cauchy data are unrestricted." Positive-order *scalar* coefficients in the Einstein jet have the X-A zero Cauchy data (XI-A def:jet), but the free background (γ_0, ϕ_0) is unrestricted. Thus $\phi = x^1$ and $h_{12} = p(t - z)$ (smooth free solutions of $\Box\phi = 0$ and $G_{\mathrm{lin}} h = 0$) are admitted by the frozen contract. Route A is therefore available and sufficient for the massless theorem: the displayed fields of Proposition 9 qualify. Hypothesis 9.6 is discharged by Route A alone.

Route B (spatially compact Cauchy data; not claimed discharged). A previous draft proposed domain-of-dependence cutoffs relative to $\overline{\operatorname{supp} f \cup \operatorname{supp}\xi \cup \operatorname{supp}(g - \sigma) \cup U}$. That set is *not* spatially compact: for $t > \varepsilon$ one has $\sigma = 1$, while $g = 0$ outside the compact spatial support of $\kappa_0\mu$, so $g - \sigma = -1$ on an unbounded spatial region. Cauchy shadows of $\operatorname{supp}(g - \sigma)$ are therefore not spatially compact, and the draft construction is invalid as written.

A repair would replace $\operatorname{supp}(g - \sigma)$ by an explicitly identified *compact* region K_{rel} that actually enters the seed and root pairing (e.g. a closed neighbourhood of $\overline{U}$ together with the compact cone-meeting set in N from Lemma 8, on which $\alpha = -S\,\mathfrak{a}^{\mu 1}\partial_\mu\psi$ is already supported), and prove that free solutions agreeing

with $(\phi, h) = (x^1, p(t-z))$ on $K_{\rm rel}$ preserve $H|_U$ and $s_0 F_t$. That repair is **not** carried out in this revision. Route B is therefore **not claimed discharged**. The massless obstruction uses Route A only.

The algebraic gravity gadget below remains on record as a constraint-compatible TT generator (certificate `onshell_admission_certificates.json`), but it does not by itself discharge Route B without the compact-region repair just described. **Gravity gadget (not a Route B discharge).** Let $K_{\mu\alpha\nu\beta}$ be the constant 4-tensor

$$K = \tfrac{1}{2}\big((e_1 \wedge e_0) \otimes (e_2 \wedge e_0) + (e_2 \wedge e_0) \otimes (e_1 \wedge e_0)\big)$$

(Riemann symmetries). For any scalar Φ, set $Z_{\mu\alpha\nu\beta} = \Phi\, K_{\mu\alpha\nu\beta}$ and $\bar{h}_{\mu\nu} = \partial^\alpha \partial^\beta Z_{\mu\alpha\nu\beta}$. Then $\partial^\mu \bar{h}_{\mu\nu} = 0$ identically, and $G_{\rm lin}(P\bar{h}) = 0$ whenever $\Box\Phi = 0$ (all ten components). For $\Phi = F(t-z)$, one recovers the TT plane wave $h_{12} = p(t-z)$ upon choosing $F'' = 2p$.

Lemma 10 (Field-only projection)**.** *Let*

$$A_0 = \int \big(h^{*\mu\nu} U_{\mu\nu} + \phi^* V + c^*_\mu W^\mu\big)$$

be antifield-linear of ghost numbers $\mathrm{gh}(U, V, W) = (1, 1, 2)$*, with coefficients U, V, W regular multilocal functionals of the fields (no antifields) in the Gate N-Ins kernel class. Write P_0 for the projection onto antifield number zero (field-only part) and $Q_0 = (S_0, \cdot)$ for the free differential of the minimal free extended action*

$$S_0 = \tfrac{1}{2}\langle h, Ah\rangle - \tfrac{1}{2}\langle \phi, (\Box + m^2)\phi\rangle + \int h^{*\mu\nu} R_0(c)_{\mu\nu}$$

(XI-B: the free scalar Hessian is $-(\Box + m^2)$). Then, under the frozen antibracket of Section 2,

$$P_0 Q_0 A_0 = \big\langle Ah, U\big\rangle - \big\langle (\Box + m^2)\phi, V\big\rangle = \big\langle E_h^{(0)}, U\big\rangle + \big\langle E_\phi^{(0)}, V\big\rangle, \tag{33}$$

where $E_h^{(0)} = \delta S_0/\delta h = Ah$ and $E_\phi^{(0)} = \delta S_0/\delta\phi = -(\Box + m^2)\phi$. In particular

$$P_0 Q_0 A_0\big|_{\text{free shell}} = 0. \tag{34}$$

Proof. Write the frozen antibracket $(F, G) = \sum_A \big(\frac{\delta^R F}{\delta \Phi^A}\frac{\delta^L G}{\delta \Phi^*_A} - \frac{\delta^R F}{\delta \Phi^*_A}\frac{\delta^L G}{\delta \Phi^A}\big)$. Expanding $Q_0 A_0 = (S_0, A_0)$:

- The terms $\int h^* U$ and $\int \phi^* V$ contribute, through the antifield derivatives of A_0 paired against the free Euler operators of S_0, the pairings $\langle Ah, U\rangle + \langle E_\phi^{(0)}, V\rangle = \langle Ah, U\rangle - \langle (\Box + m^2)\phi, V\rangle$, plus antifield-linear remainders coming from Q_0 acting on the field dependence of U, V. Those remainders have antifield number ≥ 1 and are discarded by P_0.
- The term $\int c^* W$ contributes through $(\delta^R S_0/\delta c)\,(\delta^L A_0/\delta c^*)$. The free minimal sector has $\delta S_0/\delta c$ containing h^* (from $\int h^* R_0(c)$), so this product has antifield number ≥ 1 and is killed by the antifield-zero projection P_0. (It is *not* claimed that S_0 has no c^* dependence in every extended model; the killing mechanism is the antifield content of $\delta S_0/\delta c$.)
- Permitted kernel operations (compactly supported multilocal products, pullbacks, and constant-coefficient differential operators already present in the inherited root) preserve the form of A_0 and the domain of the pairings, so they preserve (33).

The pairings are defined on admitted configurations 9.6 (Route A) because U, V have compact support in the observation variables and the free Euler operators are continuous in the smooth topology. Coefficients remain regular on the free shell by the regularity hypothesis of the class. Exactly one antifield appears in each summand of A_0, so antifield number is well-defined and P_0 is the stated projection. Restricting to $Ah = 0$ and $(\Box + m^2)\phi = 0$ gives (34). □

Certificate. A graded (Grassmann) exact check on a three-site lattice truncation of the minimal sector, with generic polynomial coefficients U, V, W and a nontrivial free-shell kernel, confirms (33)–(34) with the corrected scalar sign $E_\phi = -M\phi$, the vanishing of the field-only part of the c^*W contribution via the h^*-content of $\delta S_0/\delta c$, the orientation of the frozen bracket, and a negative control with the reversed bracket (`classical_bv/field_only_projection_certificates.json`). The continuum statement is Lemma 10; the lattice check is corroboration of the graded algebra, not a substitute for the continuum proof.

Theorem 11 (Conditional massless obstruction)**.** *Assume hypotheses 9.6–9.6, Lemma 8, Section 9.6, and Lemma 10. For the specified massless special-family predecessor construction of Proposition 9, no regular, seed-preserving, antifield-linear completion*

$$\mathcal{I}_t = F_t + A_0 + \kappa A_1$$

satisfies $Q_{\mathrm{inv}}\mathcal{I}_t = O(\kappa^2)$.

Proof. On the witness of Proposition 9, with $p' > 0$ on $\operatorname{supp}\xi$ and $0 \le \xi^2 \not\equiv 0$,

$$s_0 F_t = 4m_0 \int p'(t-z)\,\xi^2\, d^4x > 0.$$

Closure at field-only degree and order κ^0 requires $s_0F_t + P_0Q_0A_0 = 0$. Lemma 10 gives $P_0Q_0A_0|_{\text{free shell}} = 0$, so $s_0F_t = 0$, a contradiction. The higher-order term A_1 cannot repair a contradiction already present at order κ^0. □

Scope of Theorem 11. The theorem covers the constructed massless special-family instance only. It does not claim obstruction for every admissible kernel, nor for a fixed $m > 0$ instance without further work. It does rule out a universal completion prescription covering this instance. Hypotheses 9.6–9.6, Lemma 8, Route A of Section 9.6, and Lemma 10 are discharged in this revision (Route B is not claimed discharged); the inherited reduction identities 9.6 remain imported (re-checked pointwise, not re-proved).

Mass-dependence audit.

Object	Mass dependence	Source
Scalar operator	$P_m = \Box + m^2$, $m \ge 0$ fixed per instance	XII-D D0, XI-A
v_f defining inverse	$v_f = E^{\mathrm{A}}_m f$ (massive)	XII-D after eq:alpha
Coordinate inverse in α_{co}	E^{A}_0 (massless)	XII-D eq:alpha, E^{A}_0
Tensor root inverse	E^{A}_0 (massless); $a = -\vartheta E^{\mathrm{A}}_0 \mathcal{P}\alpha$	XII-D eq:beta, eq:compact-root
α_{sc} mass term	$-\frac{m^2}{2} g\,\eta^{ab} v_f \phi$	XII-D eq:alpha
$\mathcal{J}_{\mathrm{D}}$, A, H	mass-independent (gravitational)	XII-C / XI-A
Kernel t_{D}, $c_{\alpha\beta}$	independent of m at fixed $B_{f,g}$	XVI-I eq:push
$B_{f,g}$ through α	inherits m via v_f and the m^2 term	XII-D

Do not replace every E^{A} by E^{A}_m. The Green operator must invert the differential operator it accompanies. In particular the tensor root and the coordinate completion remain massless; only the scalar leg v_f and the explicit m^2 term in α_{sc} carry the mass. For the Track B2 deformation one may choose the observation $f_m = (\Box + m^2)\psi$ so that $v_{f_m} = E^{\mathrm{A}}_m f_m = \psi$ identically (unique advanced solution), eliminating every massive Green operator from the pairing while retaining the explicit m^2 term and the scalar deformation $\phi_m = x^1 \cos(m(t - t_*))$.

9.7 Small-mass persistence at fixed geometry

Instance family (frozen geometry; varying f_m). Keep h, ξ, ψ, g, σ, and the test region U fixed. For $m \ge 0$ set

$$f_m = (\Box + m^2)\psi, \qquad \phi_m = x^1 \cos\big(m(t - 5/2)\big), \qquad v_{f_m} = E^{\mathrm{A}}_m f_m = \psi.$$

Thus ψ solves the inhomogeneous massive equation with source f_m, so the predecessor kernel instance (through f_m) *varies with mass* while the metric / ghost / cutoff data remain frozen. Define

$$Y := A(\mathcal{L}_\xi h), \qquad W(m) := -\langle a_m, Y\rangle,$$

where a_m is the advanced field built from the special-family source α_m at mass m (XII-D / Prop. 9 at $m = 0$).

Exact source decomposition. Write $\delta\phi_m := \phi_m - x^1$ and

$$\Sigma_\mu := f_m \partial_\mu \phi_m - \Box(\psi \partial_\mu \phi_m).$$

Compact support of $\psi\partial_\mu\phi_m$ yields

$$E_0^{\mathrm{A}}(f_m \partial_\mu \phi_m) = \psi \partial_\mu \phi_m + E_0^{\mathrm{A}} \Sigma_\mu.$$

On a neighbourhood of the local source support one has $g = 1 - S$ (Lemma 8: $\kappa_0 = \mu = 1$ there), and $\sigma = 1$ throughout the causal region entering the pairing with Y ($t \geq \varepsilon$). Matching XII-D $\alpha_{\mathrm{co}} = -\sigma\, \mathfrak{a}^{\lambda\mu,ab} \partial_\lambda E_0^{\mathrm{A}}(f \partial_\mu \phi)$ (and the script densities), the difference admits

$$\alpha_m - \alpha_0 = \delta\alpha_{\mathrm{loc}} + \delta\alpha_{\mathrm{nl}}$$

with

$$(\delta\alpha_{\mathrm{loc}})_{ab} = -\sum_{\lambda,\mu} \mathfrak{a}^{\lambda\mu}{}_{ab}\, \partial_\lambda\big(\psi\, \partial_\mu \delta\phi_m\big) + (1-S) \sum_{\mu,\nu} \mathfrak{a}^{\mu\nu}{}_{ab}\, (\partial_\mu \psi)\, (\partial_\nu \delta\phi_m) - \tfrac{m^2}{2}(1-S)\, \eta_{ab}\, \psi\, \phi_m,$$

$$(\delta\alpha_{\mathrm{nl}})^{ab} = -\sum_{\lambda,\mu} \mathfrak{a}^{\lambda\mu,ab}\, \partial_\lambda E_0^{\mathrm{A}} \Sigma_\mu.$$

Equivalently, with $T_\lambda^{ab} := (\mathcal{P}\mathfrak{a}^{\lambda\mu}\Sigma_\mu)^{ab}$,

$$\mathcal{P}\delta\alpha_{\mathrm{nl}} = -\sum_\lambda \partial_\lambda E_0^{\mathrm{A}} T_\lambda.$$

(The derivative and overall minus are required: without them the subsequent integration by parts would not produce the *positive* double-Green pairing recorded below.)

Pairing identity. Integration by parts against the fixed test Y (root-adjoint / XII-D orientation) gives

$$W(m) - W(0) = \langle E_0^{\mathrm{A}} \mathcal{P}\delta\alpha_{\mathrm{loc}}, Y\rangle + \sum_\lambda \langle (E_0^{\mathrm{A}})^2 T_\lambda, \partial_\lambda Y\rangle.$$

(The second sum is the positive double-Green term obtained from $\mathcal{P}\delta\alpha_{\mathrm{nl}} = -\sum_\lambda \partial_\lambda E_0^{\mathrm{A}} T_\lambda$.)

Cone constants K_1, K_2. On the causal window used by the certificate, advanced cones from $\overline{U}$ meet the time slab $\operatorname{supp} S = [2,3]$ only at spatial radii in $[0.726, 2.274]$ (Lemma 8). Write $\Phi(t)$ for the squared radial measure of the $t =$ const sphere slice of the advanced cone that can meet the source slab for the frozen geometry,

$$\Phi(t) = \tfrac{1}{2}\Big((3.2-t)^2 - (1.8-t)^2\Big) = \tfrac{1}{2}\big((32/10 - t)^2 - (18/10 - t)^2\big)$$

(when both roots are positive; truncated to the available interval otherwise). Evaluating at the earliest contributing time $t_\star = 91/100$ in the Kirchhoff window gives the cone estimate that bounds the advanced field using the *source supremum norm*, which is then paired against the test field's L^1 norm; the resulting rational factor is

$$K_1 = \Phi(91/100) = \frac{1113}{500}.$$

For the nonlocal channel the double advanced operator requires an extra radial integral against the same measure. With R the additional radius variable along the cone and $t_\star = 91/100$,

$$K_2 = \int_0^{18/10-t_\star} R\,\Phi(t_\star+R)\,dR + \int_{18/10-t_\star}^{32/10-t_\star} R\cdot\tfrac{1}{2}(32/10-t_\star-R)^2\,dR = \frac{11197151}{10000000},$$

by exact Rational antiderivatives (script `nins3_small_mass_bound.py`, cone block). These are the constants used below; they are not free parameters.

Intermediate m^2-scaled bounds. For $0 < m \le 1$ let A_{ab} be an inclusion-preserving upper bound for the *projected* local density matching the script array `PdA`,

$$A_{ab} \;\ge\; \big|(\mathcal{P}\delta\alpha_{\mathrm{loc}})^{ab}\big|/m^2,$$

and let $B_{\lambda ab}$ bound $|T^{ab}_\lambda|/m^2$. (Thus A, B already absorb one m^{-2}; they are *not* bounds on the unprojected source alone.) Then

$$\big|\langle E_0^{\mathrm{A}}\mathcal{P}\delta\alpha_{\mathrm{loc}}, Y\rangle\big| \le m^2\,K_1\sum_{a,b} A_{ab}\,\|Y^{ab}\|_{L^1},$$

$$\Big|\sum_\lambda \langle (E_0^{\mathrm{A}})^2 T_\lambda, \partial_\lambda Y\rangle\Big| \le m^2\,K_2\sum_{\lambda,a,b} B_{\lambda ab}\,\|\partial_\lambda Y^{ab}\|_{L^1}.$$

Therefore

$$|W(m)-W(0)| \le m^2\left[K_1\sum_{a,b} A_{ab}\,\|Y^{ab}\|_1 + K_2\sum_{\lambda,a,b} B_{\lambda ab}\,\|\partial_\lambda Y^{ab}\|_1\right] \qquad (0\le m\le 1).$$

Every term accumulated by the script (projected local $A_{ab}\|Y^{ab}\|_1$, nonlocal $B_{\lambda ab}\|\partial_\lambda Y^{ab}\|_1$, cone constants K_1, K_2, and the final outward C_{hi}) appears in this inequality.

Release constants (exact). The certificate `code/insertion/nins3_small_mass_bound.py` evaluates the written inequality by inclusion-preserving interval arithmetic with Rational coefficients. Set the release constants

$$C_{\mathrm{release}} = 17843, \qquad W_{\mathrm{release}} = \frac{33566}{10^8}, \qquad m_{\mathrm{release}} = \frac{13715}{10^8}.$$

Exact Rational arithmetic yields $C_{\mathrm{release}} m_{\mathrm{release}}^2 < W_{\mathrm{release}}$. The certificate verifies $C_{\mathrm{iv,upper}} \le C_{\mathrm{release}}$ and $W_{\mathrm{release}} \le W_{0,\text{certified lower}}$ (the massless enclosure lower endpoint, audited at export). Hence $W(m) > 0$ on

$$\boxed{0 \le m < 1.3715\times 10^{-4}}.$$

The raw square root $\sqrt{W_0/C_{\mathrm{hi}}}$ is recorded as a diagnostic only (no claim that ordinary floating-point square roots implement downward rounding). The bound is conditional on the same inherited reduction identities as $W(0)$. This is an obstruction on a proved mass interval, not for every prescribed $m > 0$ (Route B and arbitrary-mass remain unresolved).

Scaling audit (arbitrary prescribed mass). The frozen contract fixes the preparation data f, g, σ_ε and the observation region U for each instance (XII-D/XI-B plateau conditions; Gate N-Ins.1). It does *not* declare a free length-scale parameter L that may be varied after the instance is chosen. Rescaling to $X = (x - x_*)/L$ would change $\operatorname{supp} f$, $\operatorname{supp} g$, and the relative geometry of the g-dip and the $\sigma = 1$ plateau; whether such a change stays inside a single frozen instance is not granted by the contract.

Verdict. The localized scaling route of Track B3 is **unavailable** as an automatic extension of the present instance. A future proof that shrinking the geometry is permitted, and that a normalized constant $|W(m,L)/W(0,L) - 1| \le C(mL)^2$ stays controlled, would reopen it. Until then, arbitrary prescribed $m > 0$ remains outside the small-mass interval of Section 9.7.

9.8 N-Ins.5 — prepared source

Contingent: $(S_D, \mathcal{I}_t) = (V_\sigma + V_g, \mathcal{I}_t) + O(\kappa^2)$ only after invariant completion. Second source order excluded. Structural certificate: `insertion/source_master_identity.tex`.

10 Milestone table

Milestone	Status	Note
N-A	Constructed	odd signs, antifield prose, S_{inv}, $D_{\mathrm{div}}/D_{\mathrm{gf}}$
N-Ins.1	Constructed	baseline + completion space
N-Ins.2	Constructed	actual t_{D}, E_F, J_t
N-Ins.3	Analytic 4D nonvanishing (massless sp. fam.; conditional)	Thm. 11; $m > 0$: Sec. 9.7
N-Ins.4	Unresolved (universal)	instance obstructed; inherited chain imported
N-Ins.5	Constructed (structural)	contingent on completion

Status after the r9 critique.

Item	Assessment
Convention contradictions (antifield rule, S_{inv})	fixed
Root-adjoint algebra ($D_{\mathrm{div}}/D_{\mathrm{gf}}$)	consistent
Local massless current (w, $H\|_U$, $\mathcal{C}\|_U$)	consistent (actual compact construction)
Cutoff exclusion (Lemma 8)	discharged
Validated enclosure	reported ($s_0 F_t \in [3.3566 \times 10^{-4}, 3.3637 \times 10^{-4}]$)
Configuration-space admissibility (Sec. 9.6)	Route A discharged; Route B not claimed
Massless obstruction (Thm. 11)	conditional (Route A sufficient; imported reduction identities)
Fixed $m > 0$	certified $0 \le m < 1.3715 \times 10^{-4}$ ($C_{\mathrm{hi}} \le 17843$)
Antifield-linear completion (universal)	Unresolved
N-A release	Constructed

Non-claim 12. No UV; no loops; no graph closure; no singular extensions; toy-kernel and inadmissible $(g-1)$ on ghost support withdrawn; on-shell: analytic nonvanishing for the massless special family, conditional on the inherited reduction identities; bump membership is $\operatorname{supp} S = [2,3]$, $S \in C_c^\infty(\mathbb{R})$; 1+1 is a model variation, not an obstruction; values for the superseded vector $E^{\mathrm{A}}(D_{\mathrm{div}}\mathcal{P}\alpha)$ withdrawn; numerical evidence is corroboration, not a certificate; sampling $\neq$ universal identity certificate; graph↔action vertex identification unsettled; XVI-M Unresolved normalization items beyond N-Norm$_{\mathrm{free}}$ untouched; Theorem 11 covers one constructed instance, not every kernel and not fixed $m > 0$.

11 Evidence table

Claim	Kind	Pointer
Odd S_{af} / generators	written+reported	Sec. 2
$(S_D, S_D) = \kappa \int c(\mathcal{B}_g - B)$	written	Prop. 5
ℓ_3, J_3, A, Q, N_0	imported+impl.	Sec. 4–5
af0 universal	imported+written	Gate 4
$s\Psi$ sign / vertices	written+regression	Sec. 7.1
t_{D} definition	imported quote	XVI-I eq:push; XVI-H thm:trep
$s_0 F_t = 2 \int a\, \mathcal{J}_D(R_0, h)$	written	(28)
$J_3(R_0, h) \not\equiv 0$ on shell	reported	onshell_structural_*
$H = \alpha + 2\mathcal{P} D^*_{\mathrm{div}} E^{\mathrm{A}} D_{\mathrm{div}} \alpha$	exact cert.	(31)

Claim	Kind	Pointer
4D $s_0 F_t \neq 0$ ($m = 0$ sp. fam.)	analytic + enclosure	Prop. 9, Lem. 8
Support / cutoff exclusion	exact	Lem. 8
Config. admission	Route A discharged; Route B not claimed	Sec. 9.6
$P_0 Q_0 A_0\|_{\text{shell}} = 0$	written + graded check	Lem. 10
Conditional massless obstruction	theorem	Thm. 11
Mass-dependence audit	written	Sec. 9.6
1+1 model variation	exact (model)	Sec. 9.4
Toy witness / "Proved obstructed"	withdrawn	removed

12 Independent-check methodology

- BRST: Cartan/Lie + unrestricted 2-jet $\mathbb{Q}$ samples; negative controls.
- Generators: Hamiltonian extraction vs table for all fields.
- af0: XII-C import + conversion; sampling corroboration only.
- t_{D}: continuum identities from quoted predecessors; no toy kernel.
- Root adjoint: independent sympy certificate $H_{\text{direct}} = 2G_{\text{lin}}(a_{\text{root}})$ vs reduced (traceful and trace-free).
- On-shell: 4D special family exact cone integrals + validated interval enclosure; 1+1 model recomputed.
- Regression: `math_passes` vs `packaging_passes` separated.

13 Release ladder

Item	Status
$\text{N-Norm}_{\text{free}}$	Constructed
Full retained-order broken master (N-A)	**Constructed**
Gate N-Ins.1–2,5	Constructed
Gate N-Ins.3	Analytic 4D nonvanishing (massless; conditional; Route A); certified $0 \leq m < 1$
Gate N-Ins.4	Unresolved
N-B–N-E	not claimed

14 Artifacts

- `conventions/convention_sheet.md`
- `insertion/factorization_P0Q0A0.md`
- `insertion/root_adjoint_H.py`, `insertion/root_adjoint_Ddiv_Dgf_check.py`, `insertion/cone_kirchhoff.py`
- `insertion/nins3_4D_special_family_analytic.py`, `insertion/nins3_4D_validated_enclosure.py`
- `insertion/nins3_1p1_corrected.py`, `insertion/nins3_4D_kirchhoff_corrected.py`
- `conventions/source_of_truth.py`
- `classical_bv/generator_antibracket_test.py`, `extended_action_master.py`, `s_psi_coefficients.py`

- prep/prepared_noether_af0.py, general_derivation_af0.md
- insertion/tD_definition.md
- insertion/seed_functional_derivatives.tex
- insertion/free_obstruction_current.tex
- insertion/onshell_structural_check.py
- insertion/admissibility_verdict.md
- insertion/parity_control_symmetric.py
- insertion/invariant_baseline_certificate.tex
- insertion/completion_space.json
- insertion/obstruction_or_completion.md
- provenance/RELEASE_MANIFEST.txt
- regression_N.py

A Regression taxonomy

Mathematical checks (identities, exit codes) are listed under `math_passes`; file/string packaging checks under `packaging_passes` in `regression_N_out.txt`.

B Hash pointers for t_{D} sources

File	sha256
XVI-I.tex (/workspace)	5c99e57543cbe65134e28810b7de605e8ded2b267e633f069aec834cc534bd80
XVI-H.tex (enter-companion/raw/tex)	1552ea91d0df41bf88b1b8e7c341bfefedfc27441233e2e7fd28bfc7112e8489
XII-D.tex (enter-companion/raw/tex)	ff1f5da45640c895ccd6603bf96c6a5dcb966e781fa391368e92789476228fd9
XVI-K.tex (/workspace)	6264bc8ec68534b8486c418596139fd20610d7f49c2f09b5113171e0be0789a8
XVI-J.tex (/workspace)	dcba2602ab4af7487ce03d3de8a7bac801ca106a036f65b9afbae77d1f699841

Full digests and property list: `insertion/tD_definition.md`.